

DSC 4804 Statistical Learning
School of Engineering & Technology (SoET)

PROJECT CHARTER

SECTION A: PROJECT IDENTIFICATION

Course Code: DSC4804

Course Name: Statistical Learning

Project Title: Statistical Evaluation and Extension of Molecule Attention Transformer (MAT) for Molecular Property Prediction using Graph Structure, 3D Geometry, and Attention-based Learning

Group Number: Group 14

Date: August 23, 2026

SECTION B: TEAM COMPOSITION

No	Reg No	Student Name	Role	Responsibility
1	230C2030225	Yash Yadav	Team Lead & DL Architect	MAT implementation, attention mechanisms, milestones
2	230C2030214	Kartavya Dev	Data Engineering & Featurization	Preprocessing, RDKit conformers, distance matrices, splits
3	230C2030243	Akshat Nagori	Statistical Learning & Baselines	RF/SVM/GCN, CV, hypothesis tests, ablations

SECTION C: PROBLEM STATEMENT

Background:

ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity) property prediction sits at the heart of modern drug discovery, since early and reliable estimates of these properties can save enormous amounts of time and money by ruling out unpromising candidate molecules before they ever reach the lab. Over the years, researchers have approached this problem from quite different angles: classical QSAR models built on ECFP fingerprints paired with Random Forest or SVM, more recent SMILES-based Transformers that treat molecules as text, Graph Neural Networks (GNNs) that operate directly on molecular graphs, and the Molecule Attention Transformer (MAT), which stands out by folding both molecular graph topology and 3D Euclidean inter-atomic distances directly into the Transformer's attention mechanism. Each of these families of models makes different trade-offs, and it is not always obvious which inductive biases actually help.

Problem Definition:

Despite all this progress, a few gaps keep showing up in the literature. Most papers do not clearly separate how much of MAT's reported performance comes from pretraining versus from the architecture itself, which makes it hard to know what is really driving the results. Many evaluations also lean on optimistic, randomly shuffled data splits that overstate how well a model would actually generalize to genuinely new molecules. On top of that, very few studies try to quantify exactly how much the 3D geometry information is contributing, and almost none explore a simple ECFP+MAT hybrid that could be especially useful when working with small, data-scarce datasets. This project sets out to statistically evaluate and extend MAT by benchmarking it head-to-head against RF, SVM/SVR, GCN, and a set of MAT ablations (MAT-NoGraph, MAT-NoDistance, MAT-NoAttention), tested across both Random (i.i.d.) and Bemis-Murcko Scaffold (OOD) splits, on six MoleculeNet datasets — BBBP, ESOL, FreeSolv, MetStab, Estrogen Alpha, and Estrogen Beta. The central hypothesis we are testing is that augmenting attention with graph structure and 3D distance information genuinely improves both prediction accuracy and out-of-distribution generalization, rather than simply adding complexity without real benefit.

Relevance and Impact:

By carrying out this kind of rigorous, side-by-side benchmarking of statistical and deep learning models, this work aims to speed up virtual screening in drug discovery and give practitioners a clearer, more trustworthy picture of which modeling choices actually matter. Understanding precisely how much graph structure and 3D geometry each contribute helps demystify why some models generalize better than others, rather than leaving that as a black box. And by demonstrating whether a hybrid ECFP+MAT approach can meaningfully help in low-data regimes, the project also points toward practical, resource-efficient strategies that smaller labs and teams working with limited datasets could realistically adopt.

SECTION D: OBJECTIVES (In Bullet Points)

Primary Objectives

- Implement and carefully tune Random Forest (RF), SVM/SVR, and GCN baselines so that comparisons against MAT are fair rather than stacked in favor of the newer model.
- Reproduce Molecule Attention Transformer (MAT) from scratch, generating 3D conformers using RDKit's ETKDG algorithm.
- Conduct a systematic set of ablation studies:
 - MAT-NoGraph
 - MAT-NoDistance
 - MAT-NoAttention
- Compare pretrained MAT vs scratch MAT to isolate exactly which components are doing the heavy lifting.
- Evaluate all models using both:
 - Random (i.i.d.) split
 - Bemis-Murcko Scaffold (OOD) split to obtain an honest assessment of generalization instead of relying on a single split.
- Build an ECFP-MAT late-fusion hybrid to evaluate whether combining classical fingerprints with learned representations improves performance.

- Analyze multi-head attention weights to understand the molecular patterns the model is actually learning.

Success Criteria (Measurable)

- Report ROC-AUC, PR-AUC, and F1-score for classification tasks.
- Report RMSE, MAE, and R^2 for regression tasks.
- Measure and report the generalization gap between Random and Scaffold splits.
- Validate model comparisons using rigorous statistical testing rather than visual inspection.
- Perform 5-fold cross-validation across five random seeds (42, 123, 456, 789, 1011).
- Use paired t-tests to evaluate statistical significance.
- Use Wilcoxon signed-rank tests as a non-parametric significance test.
- Report 95% confidence intervals for performance metrics.
- Consider improvements statistically significant only when $p \leq 0.05$.

SECTION E: METHODOLOGY

Technical Approach:

Phase	Description
I	We start with a thorough literature review, working through QSAR, SMILES Transformers, GNNs, and MAT in enough depth to write out their mathematical formulations ourselves, so that everyone on the team genuinely understands the models before we start coding them.
II	Next comes dataset curation: pulling and cleaning the six MoleculeNet datasets we are using (BBBP, ESOL, FreeSolv, MetStab, Estrogen Alpha, and Estrogen Beta), followed by generating 3D conformers for every molecule using RDKit's ETKDG algorithm so we have geometric information ready for the later modeling steps.
III	From those conformers we construct the adjacency and distance matrices that the graph- and geometry-aware models will need, and in parallel we implement the baseline models (RF, SVM/SVR, GCN) and reproduce MAT itself in PyTorch, aiming to match the architecture as faithfully as we can.
IV	With the models in place, we move into ablation experiments (MAT-NoGraph, MAT-NoDistance, MAT-NoAttention) to see what each component actually contributes, run a pretrained-vs-scratch comparison, and evaluate everything under both Random and Bemis-Murcko Scaffold splits to see how performance holds up when the test molecules look genuinely different from the training ones.
V	Finally, we bring everything together with a careful statistical evaluation, running 5-fold cross-validation across the five seeds (42,

	123, 456, 789, 1011) and applying paired t-tests, Wilcoxon signed-rank tests, and 95% confidence intervals, so that our conclusions about which model works better are backed by solid evidence rather than a single lucky run.	
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Tools and Technologies:

- Python 3.10+, PyTorch 2.1+, and PyTorch Geometric 2.4+ for implementing and training the MAT model, GCNs, and other deep learning components.
- RDKit 2023.09+ for cheminformatics tasks, including molecular parsing, feature extraction, and ETKDG-based 3D conformer generation.
- Scikit-Learn 1.3+, NumPy, Pandas, and SciPy for classical baseline models, data preprocessing, numerical computation, and statistical testing.
- Matplotlib and Seaborn for visualizing model performance, attention analysis, and experimental results.
- Jupyter Notebooks, CUDA-enabled GPUs, and Git/GitHub for experimentation, accelerated training, reproducibility, version control, and team collaboration.

SECTION F: TEAM AGREEMENT

We, the undersigned team members, agree to:

- ☒ Participate equally and fulfill assigned responsibilities
- ☒ Attend all scheduled meetings and discussion sessions
- ☒ Maintain honest and complete documentation
- ☒ Disclose all AI tool usage transparently
- ☒ Complete individual viva and demonstrate understanding
- ☒ Not engage in plagiarism or academic misconduct
- ☒ Resolve conflicts professionally through discussion or faculty guidance

Consequences of non-compliance: As per institutional academic regulations

SECTION G: SIGNATURES

Student 1: Yash Yadav

Signature: 

Date: 23/08/2026

Student 2: Kartavya Dev

Signature: 

Date: 23/08/2026

Student 3: Akshat Nagori

Signature: 

Date: 23/08/2026